

WEEK 1

SIMPLE COMMANDS

Topic 1: Basic Linux Commands

```
[eeshayuktha@Eeshas-MacBook-Air ~ % pwd
/Users/eeshayuktha
[eeshayuktha@Eeshas-MacBook-Air ~ % whoami
eeshayuktha
[eeshayuktha@Eeshas-MacBook-Air ~ % date
Tue Aug  4 21:04:58 IST 2026
[eeshayuktha@Eeshas-MacBook-Air ~ % history
 79 gcc prun.c -o prun
 80 nano prun.c
 81 gcc prun.c -o prun
 82 ./prun
 83 nano baexec
 84 nano baexec.c
 85 gcc baexec.c -o ba
 86 nano baexec.c
 87 gcc baexec.c -o ba
 88 ./ba
 89 nano childcomp.c
 90 gcc childcomp.c -o cc
 91 ./cc
 92 pwd
 93 clear
 94 pwd
[eeshayuktha@Eeshas-MacBook-Air ~ % cal
    August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
[eeshayuktha@Eeshas-MacBook-Air ~ % hostname
Eeshas-MacBook-Air.local
[eeshayuktha@Eeshas-MacBook-Air ~ % uname -a
Darwin Eeshas-MacBook-Air.local 25.5.0 Darwin Kernel Version 25.5.0: Tue Jun  9 22:27:52 PDT 2026; root:xnu-12377.121.10
~1/RELEASE_ARM64_T8112 arm64
```

Topic 2: Directory Management

```
eeshayuktha@Eeshas-MacBook-Air ~ % ls
ba      Desktop      hello      OS          pid          prun          SampleProgram.c
baexec  Documents     Library    os1         pid.c        prun.c        un
baexec.c Downloads     Movies     pa          printname.c  Public
cc      fork          Music      pa.c        pro          readUsername
childcomp.c fork.c       name       Pictures    process.c    readUsername.c

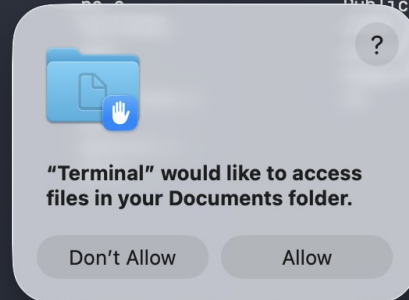
[eeshayuktha@Eeshas-MacBook-Air ~ % ls -l
total 816
-rwxr-xr-x  1 eeshayuktha  staff  33472  4 Aug 20:54 ba
-rw-r--r--  1 eeshayuktha  staff   178  4 Aug 20:53 baexec
-rw-r--r--  1 eeshayuktha  staff   161  4 Aug 20:54 baexec.c
-rwxr-xr-x  1 eeshayuktha  staff  33536  4 Aug 20:58 cc
-rw-r--r--  1 eeshayuktha  staff   265  4 Aug 20:58 childcomp.c
drwx-----+ 4 eeshayuktha  staff   128  4 Aug 22:13 Desktop
drwx-----+ 10 eeshayuktha  staff   320  3 Aug 14:09 Documents
drwx-----+ 19 eeshayuktha  staff   608  4 Aug 21:00 Downloads
-rwxr-xr-x  1 eeshayuktha  staff  33464  4 Aug 20:44 fork
-rw-r--r--  1 eeshayuktha  staff   105  4 Aug 20:44 fork.c
-rwxr-xr-x  1 eeshayuktha  staff  33432 31 Jul 14:41 hello
drwx-----@ 87 eeshayuktha  staff   2784  1 Aug 19:05 Library
drwx-----  4 eeshayuktha  staff   128 28 Jul 21:33 Movies
drwx-----+  4 eeshayuktha  staff   128  1 Aug 15:03 Music
-rwxr-xr-x  1 eeshayuktha  staff  33592  4 Aug 20:38 name
drwxr-xr-x  5 eeshayuktha  staff   160  4 Aug 19:42 OS
drwxr-xr-x  2 eeshayuktha  staff    64 31 Jul 09:36 os1
-rwxr-xr-x  1 eeshayuktha  staff  33456  4 Aug 20:48 pa
-rw-r--r--  1 eeshayuktha  staff   231  4 Aug 20:48 pa.c
drwx-----+ 5 eeshayuktha  staff   160  3 Aug 09:12 Pictures
-rwxr-xr-x  1 eeshayuktha  staff  33552  4 Aug 20:50 pid
-rw-r--r--  1 eeshayuktha  staff   289  4 Aug 20:50 pid.c
-rw-r--r--  1 eeshayuktha  staff   159  4 Aug 20:37 printname.c
-rwxr-xr-x  1 eeshayuktha  staff  33472  4 Aug 20:42 pro
-rw-r--r--  1 eeshayuktha  staff   111  4 Aug 20:42 process.c
-rwxr-xr-x  1 eeshayuktha  staff  33512  4 Aug 20:52 prun
-rw-r--r--  1 eeshayuktha  staff   260  4 Aug 20:52 prun.c
drwxr-xr-x+  5 eeshayuktha  staff   160 31 Jul 15:05 Public
```

```
eeshayuktha@Eeshas-MacBook-Air ~ % ls -la
.          .zsh_sessions      fork        pa          prun.c
..         ba                fork.c      pa.c       Public
.CFUserTextEncoding  baexec      hello      Pictures  readUsername
.copilot    baexec.c     Library    pid        readUsername.c
.DS_Store  cc           Movies     pid.c      SampleProgram.c
.Trash      childcomp.c  Music      printname.c un
.vscode     Desktop      name       pro
.vscode-shared Documents    OS         process.c
.zsh_history Downloads    os1        prun

eeshayuktha@Eeshas-MacBook-Air ~ %
```

```
Last login: Tue Aug  4 23:56:28 on ttys000
eeshayuktha@Eeshas-MacBook-Air ~ % ls -la
.          .zsh_sessions      fork        pa          prun.c
..         ba                fork.c      pa.c       Public
.CFUserTextEncoding  baexec      hello      Pictures  readUsername
.copilot    baexec.c     Library    pid        readUsername.c
.DS_Store  cc           Movies     pid.c      SampleProgram.c
.Trash      childcomp.c  Music      printname.c un
.vscode     Desktop      name       pro
.vscode-shared Documents    OS         process.c
.zsh_history Downloads    os1        prun

eeshayuktha@Eeshas-MacBook-Air ~ % cd Documents
```



```
eeshayuktha@Eeshas-MacBook-Air ~ % cd Documents
eeshayuktha@Eeshas-MacBook-Air Documents % cd ..
```

```
-rw-r--r--  1 eeshayuktha  staff   111  4 Aug 20:42 process.c
-rwxr-xr-x  1 eeshayuktha  staff 33512  4 Aug 20:52 prun
-rw-r--r--  1 eeshayuktha  staff   260  4 Aug 20:52 prun.c
drwxr-xr-x+  5 eeshayuktha  staff   160 31 Jul 15:05 Public
-rw-r--r--  1 eeshayuktha  staff   130 31 Jul 15:03 readUsername
-rw-r--r--  1 eeshayuktha  staff   159 31 Jul 15:04 readUsername.c
-rw-r--r--  1 eeshayuktha  staff    70 31 Jul 14:39 SampleProgram.c
-rwxr-xr-x  1 eeshayuktha  staff 33584 31 Jul 15:05 un
```

```

[eeshayuktha@Eeshas-MacBook-Air ~ % cd Documents
[eeshayuktha@Eeshas-MacBook-Air Documents % head prun.c
head: prun.c: No such file or directory
[eeshayuktha@Eeshas-MacBook-Air Documents % cd ~/Documents/OSLab
[eeshayuktha@Eeshas-MacBook-Air OSLab % head prun.c
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

[eeshayuktha@Eeshas-MacBook-Air OSLab % wc prun.c
   22      23     260 prun.c
[eeshayuktha@Eeshas-MacBook-Air OSLab % tail prun.c
    printf("Child Running\n");
}
else
{
    wait(NULL);

    printf("Parent Running After Child\n");
}
}
[eeshayuktha@Eeshas-MacBook-Air OSLab % clear

```

Topic 3: File Management

```

[eeshayuktha@Eeshas-MacBook-Air OSLab % cd ~/Documents/OSLab
[eeshayuktha@Eeshas-MacBook-Air OSLab % touch file1.txt
[eeshayuktha@Eeshas-MacBook-Air OSLab % cat file1.txt
[eeshayuktha@Eeshas-MacBook-Air OSLab % cp a.txt b.txt
cp: a.txt: No such file or directory
[eeshayuktha@Eeshas-MacBook-Air OSLab % mv old.txt new.txt
mv: rename old.txt to new.txt: No such file or directory
[eeshayuktha@Eeshas-MacBook-Air OSLab % rm test.txt
rm: test.txt: No such file or directory
[eeshayuktha@Eeshas-MacBook-Air OSLab % head prun.c
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

[eeshayuktha@Eeshas-MacBook-Air OSLab % tail prun.c
    printf("Child Running\n");
}
else
{
    wait(NULL);

    printf("Parent Running After Child\n");
}
}

```



```

}
[eeshayuktha@Eeshas-MacBook-Air OSLab % more prun.c
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("Child Running\n");
    }
    else
    {
        wait(NULL);

        printf("Parent Running After Child\n");
    }
}

```

```

22      20      200 printf
[eeshayuktha@Eeshas-MacBook-Air OSLab % diff pid.c prun.c
1c1
< #include<stdio.h>
---
> #include<stdio.h>
2a3
> #include<sys/wait.h>
12,13c13
<         printf("Child PID = %d\n",getpid());
<         printf("Parent PID = %d\n",getppid());
---
>         printf("Child Running\n");
17c17,19
<         printf("Parent PID = %d\n",getpid());
---
>         wait(NULL);
>
>         printf("Parent Running After Child\n");
20d21
<     return 0;
22d22
<
[eeshayuktha@Eeshas-MacBook-Air OSLab % sort prun.c

```

```
}  
[eeshayuktha@Eeshas-MacBook-Air OSLab % more prun.c  
#include<stdio.h>  
#include<unistd.h>  
#include<sys/wait.h>  
  
int main()  
{  
    pid_t pid;  
  
    pid=fork();  
  
    if(pid==0)  
    {  
        printf("Child Running\n");  
    }  
    else  
    {  
        wait(NULL);  
  
        printf("Parent Running After Child\n");  
    }  
}
```

```
eeshayuktha@Eeshas-MacBook-Air ~ % cd ~/Documents/OSLab
eeshayuktha@Eeshas-MacBook-Air OSLab % nano hello.c
```

```
[eeshayuktha@Eeshas-MacBook-Air OSLab % nano hello.c
```

File: hello.c

```
^C Cur Pos
^T To Spell
```

```
"hello.c" 8L, 78B
```

```
[eeshayuktha@Eeshas-MacBook-Air ~ % cd ~/Documents/OSLab
[eeshayuktha@Eeshas-MacBook-Air OSLab % gcc hello.c -o hello
[eeshayuktha@Eeshas-MacBook-Air OSLab % ./hello
Hello Linux
[eeshayuktha@Eeshas-MacBook-Air OSLab % which gcc
/usr/bin/gcc
[eeshayuktha@Eeshas-MacBook-Air OSLab % whereis gcc
gcc: /usr/bin/gcc
[eeshayuktha@Eeshas-MacBook-Air OSLab % man gcc
No manual entry for gcc
eeshayuktha@Eeshas-MacBook-Air OSLab % █
```


Topic 6: Process Management

```
leeshayuktha@Eeshas-MacBook-Air OSLab % ps
PID TTY          TIME CMD
55354 ttys000      0:00.06 -zsh
leeshayuktha@Eeshas-MacBook-Air OSLab % ps -ef
UID    PID  PPID  C STIME TTY          TIME CMD
0      1    0    0 Fri09AM ??        16:04.50 /sbin/launchd
0     330    1    0 Fri09AM ??        11:35.06 /usr/libexec/logd
0     332    1    0 Fri09AM ??         0:40.29 /usr/libexec/UserEventAgent (System)
0     334    1    0 Fri09AM ??        6:33.79 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/FSEvents.framework/Versions/A/Support/fseventsd
0     335    1    0 Fri09AM ??        2:24.65 /System/Library/PrivateFrameworks/MediaRemote.framework/Support/mediare
moted
0     338    1    0 Fri09AM ??         0:22.88 /usr/sbin/systemstats --daemon
278    340    1    0 Fri09AM ??         0:03.49 /System/Library/PrivateFrameworks/MobileAccessoryUpdater.framework/Supp
ort/accessoryupdaterd 120
0     342    1    0 Fri09AM ??        3:06.33 /usr/libexec/configd
0     344    1    0 Fri09AM ??        1:45.53 /System/Library/CoreServices/powerd.bundle/powerd
0     345    1    0 Fri09AM ??         0:00.08 /usr/libexec/IOMFB_bics_daemon
0     349    1    0 Fri09AM ??         0:00.78 /usr/libexec/remoted
306    353    1    0 Fri09AM ??         0:00.03 /System/Library/PrivateFrameworks/CoreSpeech.framework/corespeechd_syst
em
0     355    1    0 Fri09AM ??         0:05.70 /usr/libexec/watchdogd
0     359    1    0 Fri09AM ??        11:38.82 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mds
0     361    1    0 Fri09AM ??         0:02.13 /usr/libexec/kernelmanagerd
0     362    1    0 Fri09AM ??         0:39.51 /usr/libexec/diskarbitrationd
0     365    1    0 Fri09AM ??         0:52.52 /usr/libexec/coreduetd
0     366    1    0 Fri09AM ??         0:16.63 /usr/sbin/syslogd
0     368    1    0 Fri09AM ??         0:21.71 /usr/libexec/thermalmonitord
0     369    1    0 Fri09AM ??        2:11.47 /System/Library/PrivateFrameworks/CoreDuetContext.framework/Resources/c
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
502 55465    1    0 2:22PM ??         0:00.11 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
501 55469  1368    0 2:23PM ??         0:00.29 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Frame
work.framework/Versions/150.0.7871.187/Helpers/Google Chrome Helper (Renderer).app/Contents/MacOS/Google Chrome Helper (
Renderer) --type=renderer --metrics-client-id=cd1f28b6-fd67-4ed6-86db-c25701ab25a2 --origin-trial-disabled-features=Canv
asTextNg|WebAssemblyCustomDescriptors --lang=en-GB --num-raster-threads=4 --enable-zero-copy --enable-gpu-memory-buffer-
compositor-resources --enable-main-frame-before-activation --renderer-client-id=2256 --time-ticks-at-unix-epoch=-1785472
544391638 --launch-time-ticks=105240039137 --shared-files --metrics-shmem-handle=1752395122,r,17359459865076719066,12472
09868660531124,2097152 --field-trial-handle=1718379636,r,7249616846473335252,4840708507990571897,262144 --variations-see
d-version=20260730-170103.020000-production --pseudonymization-salt-handle=1935764596,r,9732121191943575409,647176484400
6617975,4 --trace-process-track-uuid=3190711100278282838 --seatbelt-client=392
501 55470    1    0 2:23PM ??         0:00.11 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
502 55472    1    0 2:23PM ??         0:00.12 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
502 55474    1    0 2:23PM ??         0:00.11 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
502 55475    1    0 2:23PM ??         0:00.10 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
502 55476    1    0 2:23PM ??         0:00.09 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
501 55477    1    0 2:23PM ??         0:00.09 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
502 55478    1    0 2:23PM ??         0:00.10 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDSImporterWorker -m com.apple.mdworker.shared
0 55353 52473    0 2:20PM ttys000    0:00.02 login -pf eeshayuktha
502 55354 55353    0 2:20PM ttys000    0:00.07 -zsh
0 55479 55354    0 2:23PM ttys000    0:00.01 ps -ef
leeshayuktha@Eeshas-MacBook-Air OSLab %
```

Processes: 740 total, 4 running, 736 sleeping, 3784 threads 14:24:27
 Load Avg: 3.08, 3.13, 3.00 CPU usage: 3.62% user, 4.87% sys, 91.49% idle
 SharedLibs: 307M resident, 86M data, 60M linkedit. MemRegions: 0 total, 0B resident, 0B private, 605M shared.
 PhysMem: 7537M used (1667M wired, 2838M compressor), 78M unused.
 VM: 339T vsize, 6144M framework vsize, 6213653(4) swapins, 7296568(0) swapouts.
 Networks: packets: 32275549/15G in, 3594629/816M out. Disks: 29662545/642G read, 10000248/231G written.

PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORTS	MEM	PURG	CMPRS	PGRP	PPID	STATE	BOOSTS	%CPU_ME
655	fileprovider	9.2	37:40.80	7	6	179+	16M+	0B	5456K-	655	1	sleeping	*463+[1]	1.91378
55494	top	9.0	00:02.57	1/1	0	30	8880K	0B	2160K+	55494	55354	running	*0[1]	0.00000
0	kernel_task	8.0	05:26:35	578/8	0	0	57M	0B	0B	0	0	running	0[0]	0.00000
403	WindowServer	3.9	04:10:27	20	5/1	14524	589M	256K-	277M-	403	1	sleeping	*0[1]	0.03325
637	Finder	3.3	03:23.36	7	4	790	174M	0B	155M-	637	1	sleeping	*17[6]	0.88854
607	cloudd	3.0	03:35.63	8	5	282-	14M-	0B	3760K-	607	1	sleeping	*44-[2]	0.00000
1323	routined	2.1	02:50.23	7/1	6/1	296+	19M+	0B	10M-	1323	1	running	*1+[4732+]	0.00000
15533	Google Chrom	1.7	03:04:14	23	1	212	643M	0B	594M	1368	1368	sleeping	*0[446482]	0.00000
568	com.apple.Dr	1.3	37:12.21	8	6	1412	30M	0B	4448K	568	1	sleeping	0[1]	0.00000
1403	Finder	1.2	04:25.90	7	4	593	74M	0B	57M-	1403	1	sleeping	*0[5773]	0.16711
49682	Pages	1.0	05:37.54	10	5	762	472M-	0B-	360M+	49682	1	sleeping	*0[964]	0.00000
35841	ZoomUpdater	0.8	04:18.84	25	1	192	18M	0B	13M	35841	1	sleeping	*0[1]	0.00000
1027	cfprefsd	0.8	02:45.85	3	2	508+	4432K	0B	1824K-	1027	1	sleeping	*72[10301]	0.00000
334	fseventsd	0.8	06:34.30	18	1	260	5552K	0B	2432K-	334	1	sleeping	*0[1]	0.00000
359	mds	0.7	11:39.38	8	5	560	49M-	0B	41M-	359	1	sleeping	*0[1]	0.08160
689	nsurlsession	0.7	01:28.19	8	5	145-	7825K	0B	1616K	689	1	sleeping	*463[387]	0.37190
1462	mapssyncd	0.7	00:18.34	4	3	88+	7168K+	0B	3808K-	1462	1	sleeping	*4+[4225+]	0.00000
1410	calaccessd	0.6	00:41.50	3	2	379+	9808K+	0B	5184K-	1410	1	sleeping	*1+[5289+]	0.00000
35961	bluetoothd	0.6	04:18.50	11	5	522	12M	0B	5040K+	35961	1	sleeping	*0[1]	0.80086
35753	bird	0.6	00:14.78	6	5	169	11M-	0B-	2416K	35753	1	sleeping	*0[7126+]	0.16577
746	com.apple.Cl	0.5	00:08.23	3	2	73	5152K	0B	1536K	746	1	sleeping	*0[1]	0.00000
1389	suggestd	0.5	01:16.40	8	7/1	574+	17M+	0B	10M-	1389	1	sleeping	*0[4069+]	0.00000

